

Explicit three-piece quadratic model for accurate analytic representation of S-shaped solar cell current–voltage characteristics

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ARTICLE INFO

Keywords:

Solar cell
I–V characteristic
S-shaped characteristic
Analytical method
Parameter extraction
Curve fitting

ABSTRACT

Some interest has been recently directed towards analytic mathematical models for reproducing the current–voltage (I–V) characteristics of solar cells, mainly due to their convenient property of expressing current as an explicit function of voltage (which is not true for the traditional one-diode model). However, as is the case with the one-diode model itself, existing analytic models do not yield a good fit in the case of S-shaped I–V characteristics. In this brief paper, an analytic explicit model based on a three-piece quadratic curve is proposed that surpasses such difficulty and enables accurate representation of S-shaped curves. To evaluate the proposed model, validation was carried out against three empirical solar cell I–V characteristics. In all cases, simulation results show that the proposed approach is more accurate than other closed-form models in the literature.

1. Introduction

The frequently-used one-diode model (ODM) for solar cell current–voltage (I–V) characteristics is of great relevance due to its accuracy and physically meaningful parameters (Tian et al., 2012; De Soto et al., 2006). This latter aspect, in particular, makes its use essential for carrying out solar cell design computations (Messenger and Abtahi, 2017).

On the other hand, relevant criticisms with regard to the practicality of ODM in higher-level applications (e.g., design of circuits with solar cells as components) have been put forward. It is argued that the implicit nature of ODM and its nontrivial parameter extraction problem (Humada et al., 2020) can be inconvenient in such scenarios, in which it would be preferable to use analytic (i.e., explicit) models whose parameters can be readily extracted via closed-form equations, even if such parameters do not have direct physical meanings (Pindado et al., 2018).

In this context, analytic I–V models have been proposed (Das, 2013, 2011; Saetre et al., 2011; Akbaba and Alattawi, 1995; Pindado and Cubas, 2017; Karmalkar and Haneefa, 2008) whose parameters can be determined explicitly in terms of the remarkable operating points, namely those associated to operation of the solar cell at short circuit, open circuit and maximum power (Villalva et al., 2009). Such feature is relevant in practice because voltages and currents at the remarkable points are the main parameters supplied in solar cell datasheets (Pindado et al., 2018).

Corrêa and Vieira (2021) have recently proposed a new model of the aforementioned type. It is termed as explicit two-piece quadratic model (ETPQM) and was shown to provide a more consistent performance than the best models among those proposed by Das (2013, 2011), Saetre et al. (2011), Akbaba and Alattawi (1995), Pindado and Cubas (2017), Karmalkar and Haneefa (2008). In fact, for all solar cells evaluated by Corrêa and Vieira (2021), it was either the most accurate or second-best model, whereas the relative performance rankings of the other tested models varied significantly between solar cells.

Nevertheless, further evaluations carried out by Corrêa and Vieira (2022) have shown that the ETPQM shares an important problem with the ODM: it yields poor-quality fits to S-shaped I–V characteristics, which usually occur in cells fabricated with materials (e.g., organic polymers and perovskite) that may impose considerable charge transport limitations compared to more ordinary solar cells (Saive, 2019). More precisely, it was pointed out by Corrêa and Vieira (2022) that, despite being able to outperform the ODM itself for S-shaped curves, the ETPQM still incurs significant error close to the open circuit point.

Taking the above into account, this brief paper proposes an analytic model that is able to represent cells with S-shaped characteristics with high precision. It maintains the desirable piecewise-quadratic structure of ETPQM and its parameters can be extracted via simple closed-form equations, whose computation requires only a single additional I–V sample.

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<https://doi.org/10.1016/j.solener.2023.04.028>

Received 23 January 2023; Received in revised form 21 March 2023; Accepted 13 April 2023

Available online 8 May 2023

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2. Two-piece quadratic modeling

Evaluations carried out by Corrêa and Vieira (2021) have shown that, for ordinary I–V characteristics, modeling accuracy superior to that of previous explicit models can be obtained with a two-piece quadratic curve as follows:

$$I_{\text{two-piece}}(V) = \overbrace{[aV^2 + bV + c]u(V - V_{mp})}^{g_I(V)} + \underbrace{\left[\frac{-e - \sqrt{e^2 - 4d(f - V)}}{2d} \right] u(V_{mp} - V)}_{h_I(V)=[g_V(I)]^{-1}} \quad (1)$$

where $g_V(I) = dI^2 + eI + f$, $u(\cdot)$ is the unit step function, V_{mp} is the maximum power voltage, a, b, c, d, e, f are the model parameters and voltage $V \in [0, V_{oc}]$, where V_{oc} is the open circuit voltage. Let I_{mp} and I_{sc} be the maximum power current and short circuit current. The model parameters are computed according to the following equations:

$$a = \frac{I_{mp}}{(V_{oc} - V_{mp})^2} \cdot \left(\frac{V_{oc}}{V_{mp}} - 2 \right) \quad (2)$$

$$b = -\frac{2V_{mp}I_{mp}}{(V_{oc} - V_{mp})^2} \cdot \left(\frac{V_{oc}}{V_{mp}} - 2 \right) - \frac{I_{mp}}{V_{mp}} \quad (3)$$

$$c = \frac{I_{mp}V_{oc}}{V_{mp}} - \frac{V_{oc}I_{mp}(V_{oc} - 2V_{mp})^2}{V_{mp}(V_{oc} - V_{mp})^2} \quad (4)$$

$$d = -\frac{V_{mp}(2I_{mp} - I_{sc})}{I_{mp}(I_{mp} - I_{sc})^2} \quad (5)$$

$$e = \frac{2V_{mp}(2I_{mp} - I_{sc})}{(I_{mp} - I_{sc})^2} - \frac{V_{mp}}{I_{mp}} \quad (6)$$

$$f = \frac{V_{mp}I_{sc}(2I_{sc} - 3I_{mp})}{(I_{mp} - I_{sc})^2} \quad (7)$$

It is seen from (1) that such a two-piece model, which is called ETPQM (Corrêa and Vieira, 2021), fits two quadratic curves in inverse coordinate systems (i.e., I–V and V–I) and complementary domains (i.e., $[0, V_{mp}]$ and $[V_{mp}, V_{oc}]$) in the V -axis. Such approach has high accuracy because the tasks of fitting the voltage source and current source regions of the I–V characteristic are split between both quadratics.

As shown in (2)–(7), model parameters are determined solely in terms of V_{mp} , V_{oc} , I_{mp} and I_{sc} , as is usually the case with analytic I–V models (Pindado et al., 2018).

3. Proposed three-piece model

For better understanding, consider at first the scheme presented in Fig. 1, where an attempt of fitting the ETPQM to an S-shaped characteristic is illustrated. Clearly, the model is deficient at fitting the I–V characteristic for higher values of voltage. This is explained by the single parabola $g_I(V)$ not being able to approximate the S-shaped curve in the entire range $V \in [V_{mp}, V_{oc}]$, thereby resulting in large error.

Taking this into account, we now propose and develop an analytic model with the following general form:

$$I(V) = w_I(V)u(V - V_x) + I_{\text{two-piece}}(V)u(V_x - V) \quad (8)$$

where $w_I(V) = rV^2 + sV + t$ and $V_x \in]V_{mp}, V_{oc}[$ is a voltage threshold whose value will be determined in Section 4.2.

In other words, it is proposed that a third quadratic segment be fitted for $V > V_x$. By combining (1) and (8), the following analytic expression is obtained:

$$I(V) = [rV^2 + sV + t]u(V - V_x) + \left[\frac{-e - \sqrt{e^2 - 4d(f - V)}}{2d} \right] u(V_{mp} - V) + [aV^2 + bV + c] \Pi \left(\frac{V}{V_x - V_{mp}} + \frac{V_x + V_{mp}}{2} \right) \quad (9)$$

where $\Pi(\cdot)$ is the rectangular pulse function and parameters a, b, c, d, e, f are computed as per (2)–(7). In what follows, we derive closed-form expressions for extracting r, s, t and computing the voltage threshold V_x .

4. Parameter extraction

The main assumption of this work is that, aside from the remarkable operating points, *one* additional point (V_a, I_a) from the I–V curve is available, with $V_a \in]V_{mp}, V_{oc}[$. We argue that this is not an appreciable disadvantage, since sampling a *single* point either in practice or from a datasheet is quite straightforward, whereas the same cannot be said of sampling the entire curve, as is done in optimization-based parameter extraction (Humada et al., 2020).

4.1. Computing r, s and t

Given (V_a, I_a) , we now extract parameters r, s and t . The corresponding derivation is given in Proposition 1.

Proposition 1. The parameters r, s and t in (9) are given, in terms of V_a, I_a and V_{oc} , by the following equations:

$$r = \frac{I_a}{(V_a - V_{oc})^2} \quad (10)$$

$$s = -\frac{2I_aV_{oc}}{(V_a - V_{oc})^2} \quad (11)$$

$$t = \frac{I_aV_{oc}^2}{(V_a - V_{oc})^2} \quad (12)$$

Proof. We begin by enforcing $I(V) \geq 0$ for $V \in [V_a, V_{oc}]$, with equality if and only if $V = V_{oc}$. To achieve this, it is sufficient to make V_{oc} be a double root of $w_I(V)$, which yields the equations $s^2 - 4rt = 0$ and $-s/2r = V_{oc}$. At last, a third equation is obtained by using $I(V_a) = I_a$. Hence, the following nonlinear system is obtained:

$$s^2 - 4rt = 0 \quad (13)$$

$$\frac{s}{r} = -2V_{oc} \quad (14)$$

$$rV_a^2 + sV_a + t = I_a \quad (15)$$

To solve the above system, first substitute (14) into (13), from which $r(rV_{oc}^2 - t) = 0$ is easily obtained. From (14), $r \neq 0$ and thus $t = rV_{oc}^2$. Substituting this relation and (14) into (15), we are able to compute r , which is then used to obtain s and t , thereby completing the proof. ■

4.2. Determining V_x

It remains to compute the voltage threshold V_x , which is done in the proof of Proposition 2 below.

Proposition 2. The value of V_x is given by:

$$V_x = \min \left\{ \frac{b - s - \sqrt{(b - s)^2 - 4(a - r)(c - t)}}{2(a - r)}, \frac{b - s + \sqrt{(b - s)^2 - 4(a - r)(c - t)}}{2(a - r)} \right\} \quad (16)$$

where a, b, c, r, s, t are given by (2)–(4) and (10)–(12); additionally, $V_x \in \mathbb{R}$ is always satisfied.

Proof. First, note that (10)–(12) do not ensure that $w_I(V_a) = g_I(V_a)$. Hence, in general, simply using $V_x = V_a$ would make $I(V)$ discontinuous at $V = V_x$, which is not allowable. Hence, V_x must be computed as the voltage value in which $w_I(V)$ and $g_I(V)$ intersect, thereby enforcing

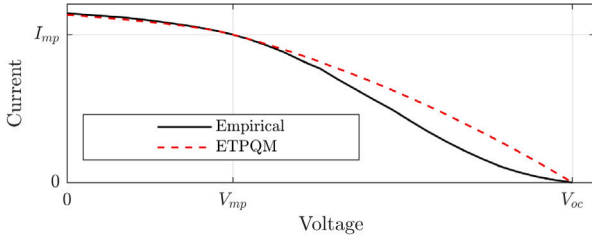


Fig. 1. Depiction of large ETPQM error for S-shaped curves.

continuity of $I(V)$. In other words, V_x is a root of the equation $w_I(V) = g_I(V)$, whose development leads to:

$$(a - r)V^2 + (b - s)V + (c - t) = 0 \quad (17)$$

which is a simple quadratic that necessarily has V_{oc} as a solution, since $g_I(V_{oc}) = 0$ (Corrêa and Vieira, 2021) and $w_I(V_{oc}) = 0$. Hence, we see that $V_x < V_{oc}$ is the smallest solution of (17), from which (16) is obtained. At last, since V_{oc} is known to be a real number and (17) has real coefficients, it is always true that $V_x \in \mathbb{R}$. ■

4.3. On the bounds of V_x

It is claimed in Section 3 that $V_x \in]V_{mp}, V_{oc}[$, which must still be proved. Consider, at first, why such bounds are *desirable*. Notice that, if a $V_x < V_{mp}$ had been considered, the combination of (1) and (8) would lead to the equation $I(V) = w_I(V)u(V - V_x) + h_I(V)u(V_x - V)$ instead of (9). This not only implies that $g_I(V)$ would bear no contribution in $I(V)$ but, even worse, that (V_{mp}, I_{mp}) would not belong to $I(V)$ because $w_I(V_{mp}) \neq I_{mp}$ in general. On the other hand, by making $V_x > V_{mp}$, it is ensured that $g_I(V)$ contributes to the I-V curve and that (V_{mp}, I_{mp}) is in $I(V)$ precisely due to the fact that $g_I(V_{mp}) = I_{mp}$ (Corrêa and Vieira, 2021).

Now, we show that the desired condition $V_x \in]V_{mp}, V_{oc}[$ is *always satisfied* if a sample voltage $V_a \in]V_{mp}, V_{oc}[$ is selected, as has been assumed in Section 4. The proof is carried out in the following Proposition 3.

Proposition 3. If $V_a \in]V_{mp}, V_{oc}[$, then $V_x \in]V_{mp}, V_{oc}[$.

Proof. At first, notice that (i) if $V_a = V_{mp}$, then $V_x = V_{mp}$ because $g_I(V_a) = w_I(V_a) = I_{mp}$ in this case and V_x is the smallest root of $g_I(V) = w_I(V)$ (see Proposition 2). Furthermore, it is clear that (ii) $\lim_{V_a \rightarrow V_{oc}} V_x = V_{oc}$. At last, (iii) V_x is a continuous function of V_a , since $g_I(V)$ and $w_I(V)$ are continuous. Combining (i)-(iii), it is seen that $V_a \in]V_{mp}, V_{oc}[$ implies that $V_x \in]V_{mp}, V_{oc}[$ necessarily. ■

5. Validation

To validate the proposed model, it is applied to three cells with S-shaped curves, whose empirical I-V data are reported by Ecker et al. (2012), Burgelman et al. (2005), Zhong et al. (2021), and its performance is compared to those of ETPQM and the power law model of Karmalkar and Haneefa (2008).

The latter model has been selected because previous evaluations have shown that, aside from ETPQM, it is the one that most often performs best for ordinary I-V curves (Pindado et al., 2018; Corrêa and Vieira, 2021). A brief description of the considered S-shaped solar cells follows:

- Solar cell 1: organic, $V_{oc} = 0.32$ V, $I_{sc} = 5.86$ mA/cm², I-V curve is reported in Fig. 1 of Ecker et al. (2012);
- Solar cell 2: CdTe, $V_{oc} \approx 0.7$ V, $I_{sc} \approx 20$ mA/cm², I-V curve is reported in Fig. 5 of Burgelman et al. (2005);

- Solar cell 3: perovskite, $V_{oc} = 1.21$ V, $I_{sc} = 17.14$ mA/cm², I-V curve is reported in Fig. 1 of Zhong et al. (2021).

The performances of the considered models are assessed via computation of normalized absolute error (NAE) and normalized root mean square error (NRMSE), which are henceforth denoted as ξ and Ξ , respectively. Such error metrics are computed with respect to points sampled from the empirical I-V characteristics and are given by:

$$\xi(V_i) = \frac{1}{I_{sc}} \cdot |I(V_i) - I_{ref}(V_i)| \quad (18)$$

$$\Xi = \left[\frac{1}{N_v} \sum_{i=1}^{N_v} \xi^2(V_i) \right]^{1/2} \quad (19)$$

where V_i , $i = 1, 2, \dots, N_v$, is the i th voltage sample of the empirical I-V curve and $I_{ref}(V_i)$ is its associated empirical current value. For each solar cell, $N_v = 50$ points with approximately equal voltage spacings were graphically sampled from its respective empirical I-V characteristic.

When applying the proposed method, a sampled voltage $V_a = \frac{1}{2}(V_{mp} + V_{oc})$ is used. Such value has been adopted since it is simple to specify based on nominal parameters and has yielded close-to-optimal performance after experimentation with various solar cells and $V_a \in [V_{mp}, V_{oc}]$ values.

The obtained results are given in Table 1, where obtained NRMSE values are shown, and Figs. 2–4, each of which displays the plots of $I(V)$, $\xi(V)$, $P(V)$ and $\partial_V I(V)$ for one of the considered cells, where $P(V) = V \cdot I(V)$ is output power and ∂_V denotes derivative with respect to voltage.

Note that $\partial_V I(V)$ is evaluated so we can qualitatively assess whether the proposed model, despite its introduction of a derivative discontinuity at $V = V_x$, still reasonably reproduces the empirical $\partial_V I_{ref}(V)$. Finally, it should be observed that, since $I_{ref}(V)$ is discrete, its derivative values at each V_i are computed via backwards difference.

6. Discussion

In general, the results show that the proposed model significantly enhances modeling accuracy for solar cells with S-shaped I-V curves compared to ETPQM and power law. In fact, relative NRMSE reductions of 64.0% (94.5%), 63.7% (71.0%) and 65.1% (72.3%) were obtained for solar cells 1 to 3, respectively, compared to ETPQM (power law).

The relatively large errors yielded by power law can mainly be attributed to it not being a piecewise model, which renders it inflexible due to the difficulty of fitting a single curve to the current source and S-shaped voltage source regions. This can be readily seen from the obtained I-V plots: for solar cell 1, a reasonable fit is achieved for $V < V_{mp}$ at the expense of a very poor fit for $V > V_{mp}$, whereas the exact opposite problem occurs for solar cells 2 and 3.

Such issue is alleviated by the ETPQM, whose piecewise nature enables it to consistently fit the current source region with low error. However, as discussed in Section 3, its single quadratic for $V > V_{mp}$ is not enough to approximate well the S-shaped voltage source region. Correspondingly, its associated NAE plots rise significantly for $V \in [V_{mp}, V_{oc}]$. The above discussion and obtained results confirm the main hypothesis that motivated the proposed model, namely that fitting a third quadratic segment in the neighborhood of V_{oc} would reduce error via increased model flexibility in the I-V curve voltage source region. This is particularly clear from the $\xi(V)$ plots, where it is seen that expressive NAE reductions are obtained for higher voltage values.

An interesting result that particularly highlights the inadequacy of power law and ETPQM for S-shaped characteristics is that obtained for solar cell 1. It is seen that, in this case, such models yield $I < 0$ for $V < V_{oc}$, which is physically unrealistic and thus undesirable. On the other hand, the proposed model prevents this because (13) and (14) ensure that $w_I(V)$ has a minimum equal to zero at $V = V_{oc}$.

Table 1
Obtained NRMSE values (%).

Model	Solar cell 1	Solar cell 2	Solar cell 3
Power law	10.4440	7.2531	9.6858
ETPQM	1.6057	5.7984	7.6947
Proposed	0.5786	2.1040	2.6832

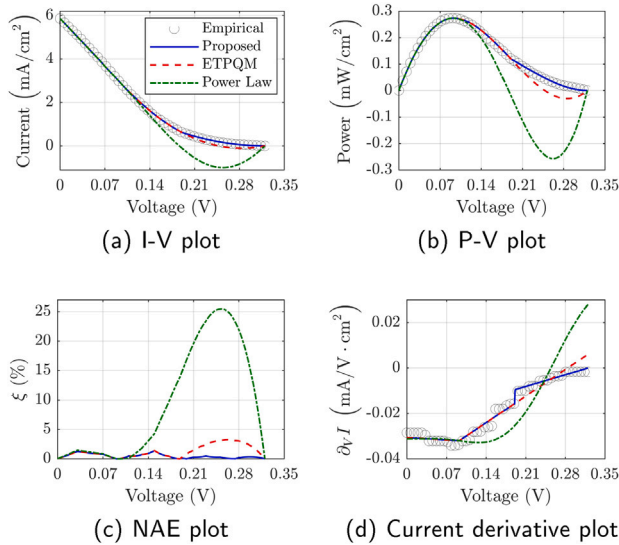


Fig. 2. Obtained results for solar cell 1.

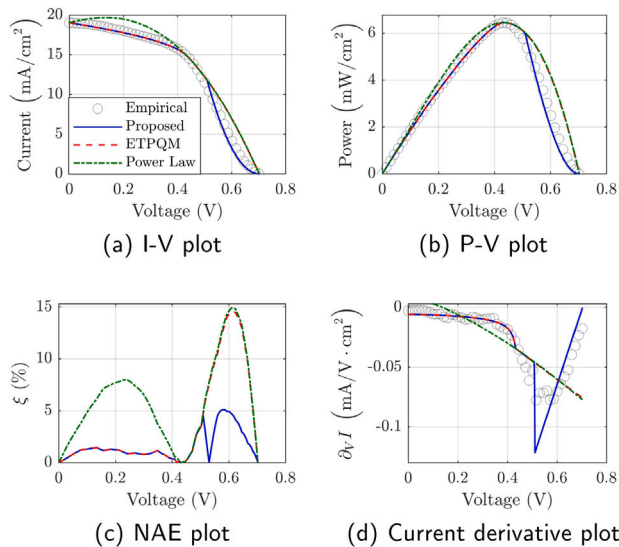


Fig. 3. Obtained results for solar cell 2.

It is noteworthy that the proposed model shares a very important feature with the other analytic models, namely that maximum power point (V_{mp} , I_{mp}) is perfectly fitted, as seen from $\xi(V_{mp}) = 0$ in the NAE plots. This is achieved because $V_x > V_{mp}$, which, with (9), implies that $I(V_{mp}) = g_I(V_{mp})$ and, in turn, it is known from Corrêa and Vieira (2021) that the quadratic $g_I(V)$ satisfies $g_I(V_{mp}) = I_{mp}$.

Finally, the current derivative plots show that some error in $\partial_V I(V)$ is indeed introduced near $V = V_x$ due to the derivative discontinuity that occurs in this point for the proposed model. However, this is compensated for by the fact that, compared to the other models, a much better fit of $\partial_V I(V)$ is obtained in the neighborhood of V_{oc} .

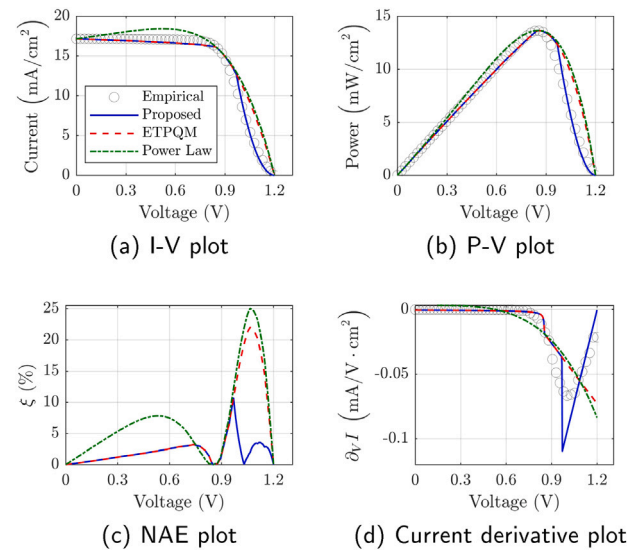


Fig. 4. Obtained results for solar cell 3.

7. Conclusion

A new analytic mathematical model for S-shaped solar cell I–V characteristics has been presented. Its proposal was motivated by the fact that the generally high-performing analytic I–V models, such as the ETPQM and power law, still lack accuracy for representing S-shaped curves. The model is highly tractable due to being piecewise-quadratic and requires only one I–V sample point, aside from the remarkable operating points, for closed-form determination of its parameters. It was validated via comparison against the power law and ETPQM approaches for three solar cells with S-shaped characteristics. Results have shown it successfully avoids the poor fit of previous models in the voltage source region, thereby yielding substantial reduction in modeling error (above 60% in all considered cases).

CRediT authorship contribution statement

Henrique Pires Corrêa: Conceptualization, Software, Methodology, Writing – original draft. **Flávio Henrique Teles Vieira:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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